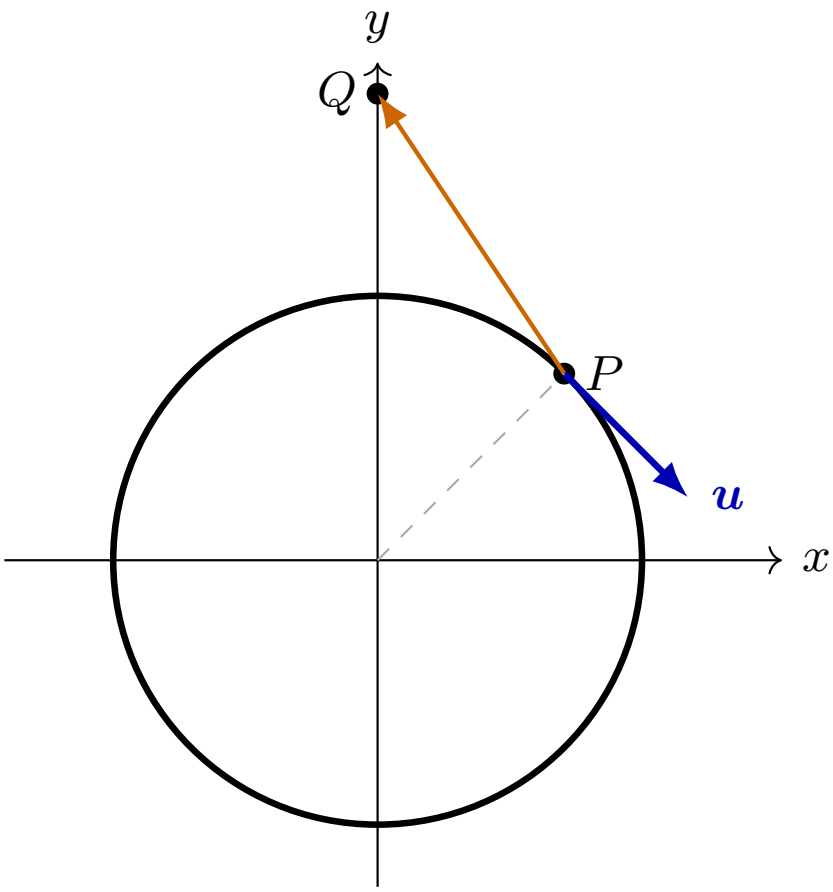
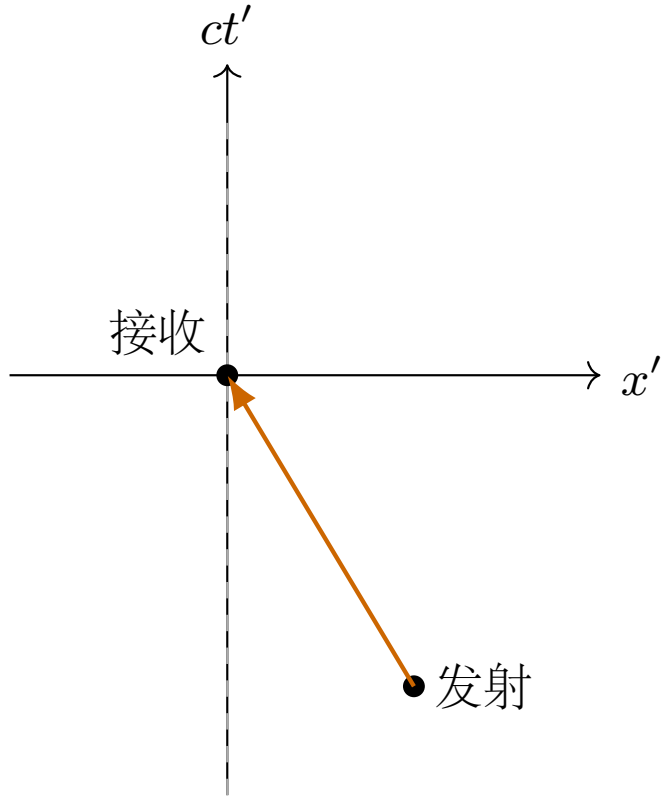


顺时针：$\boldsymbol{u} = \omega R(\sin\theta, -\cos\theta)$

光线从 P 指向 Q



接收器 $x' = 0$ 的世界线竖直

发射事件 $t'_P < 0$

图为 x' 投影，光线另有 y' 位移